

Learning goals:

- Understand covariance matrices and how to compute a covariance matrix;
- Understand the eigen decomposition of covariance matrices and how eigenvalues and eigenvectors relate to the underlying data distribution;
- Understand the limitations of estimating covariance matrices from empirical data;
- Understand shrinkage regularization;
- Understand LDA and how to implement and apply LDA;
- Understand the assumptions and limitations of LDA;
- Describe solutions to common issues of LDA.

Question 1: Refresh your understanding

Refresh your understanding in the following concepts:

- Variance and covariance: [Ben Lambert](#)
- Covariance and correlation: [Ben Lambert](#)

Question 2: Compute covariance by hand

You are given the data $\mathbf{X} \in \mathbb{R}^{N \times D}$ with $N = 3$ examples and $D = 2$ features:

$$\mathbf{X} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$$

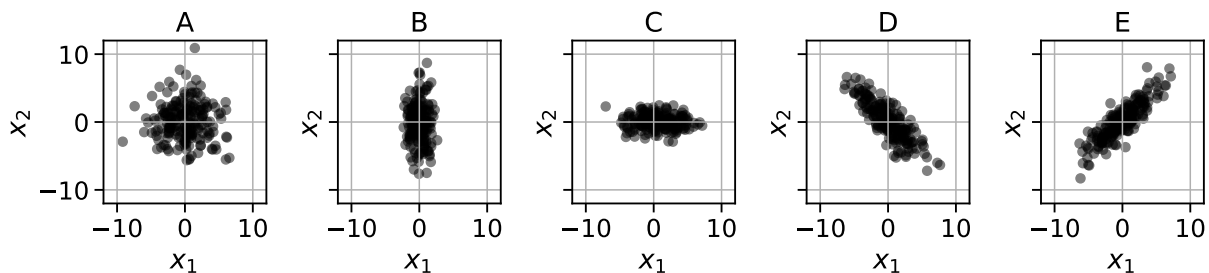
- What will be the dimensionality of the covariance matrix $\mathbf{S}$ of this dataset? That is, give $\mathbf{S} \in \mathbb{R}^{? \times ?}$.
- Compute the covariance matrix $\mathbf{S}$ by hand.

Question 3: Match covariance matrices

See the following five covariance matrices:

$$\mathbf{S}_1 = \begin{bmatrix} 8 & 0 \\ 0 & 1 \end{bmatrix}, \mathbf{S}_2 = \begin{bmatrix} 1 & 0 \\ 0 & 8 \end{bmatrix}, \mathbf{S}_3 = \begin{bmatrix} 8 & -7 \\ -7 & 8 \end{bmatrix}, \mathbf{S}_4 = \begin{bmatrix} 8 & 7 \\ 7 & 8 \end{bmatrix}, \mathbf{S}_5 = \begin{bmatrix} 8 & 0 \\ 0 & 8 \end{bmatrix}$$

Furthermore, see the following scatterplots of five zero-mean datasets:



Which covariance matrix (number) matches which dataset (letter)?

Question 4: Jupyter notebook covariance matrix

Now put the theory into practice by going through the `Covariance matrix` Jupyter notebook.

Question 5: Properties of LDA

- (a) Write down the objective function of LDA.
- (b) For each variable in the objective function, provide the dimensionality and the conceptual meaning.
- (c) Does LDA have an analytical solution?
- (d) What are the assumptions made by LDA?
- (e) Is LDA a supervised or unsupervised method?
- (f) Is LDA used for classification or for regression?

Question 6: Train an LDA

You are given the training data $\mathbf{X} \in \mathbb{R}^{N \times D}$ and class labels $\mathbf{y} \in \mathbb{N}^N$ with $N = 3$ examples and $D = 1$ features:

$$\mathbf{X} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}, \mathbf{y} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}$$

- (a) Write down the equations to learn the parameters of an LDA.
- (b) What will be the dimensionality of the weight vector? That is, give $\mathbf{w} \in \mathbb{R}^?$.
- (c) Compute the weights $\mathbf{w}$ and bias b for the provided training dataset by hand.

Question 7: Apply an LDA

You are given a trained LDA with weights $\mathbf{w} = [1, 2]$ and bias $b = 3$. Additionally, you are given the following test dataset:

$$\mathbf{X} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$$

- (a) Write down the equations to obtain the predicted class labels.
- (b) What will be the dimensionality of the output? That is, give $\hat{\mathbf{y}} \in \mathbb{N}^?$.
- (c) Compute the predicted class labels $\hat{\mathbf{y}}$ by hand.

Question 8: Apply an LDA in the context of EEG

You are given a trained LDA with weights $\mathbf{w} = [-1, 1, -1, -1, 2, -1]$ and bias $b = 2$. Additionally, you are given the following test trial of $C = 2$ channels and $T = 3$ averaged time intervals:

$$\mathbf{X} = \begin{bmatrix} 1 & 3 & 1 \\ 2 & 3 & 2 \end{bmatrix}$$

- (a) What mathematical operation is required to project the data $\mathbf{X} \in \mathbb{R}^{C \times T}$ using the weights $\mathbf{w} \in \mathbb{R}^D$?
- (b) Compute the predicted class label $\hat{y}$ by hand.

Question 9: Jupyter notebook LDA

Now put the theory into practice by going through the LDA Jupyter notebook.

Question 10: Submitting to Brightspace

For this assignment, please submit the following:

- The two Jupyter notebooks with your solutions
- A report (pdf extension) with the answers to the theoretical questions. In this report, add a line stating who contributed to what (please, adhere to the code of conduct). Additionally, please add a line on whether, where, and how you used GenAI for this assignment.

Question 11: (Optional) Test your understanding with real exam questions

- (a) Covariance matrix

- 2020-2021 Part I question b
- 2020-2021 resit Part 1 question a
- 2022-2022 resit Part 1 question d

(b) LDA

- 2020-2021 Part 4
- 2021-2022 Part 3
- 2023-2024 Part 1 question e
- 2025-2025 Part 3